🚦 Scratch Maze-Solving Robot Challenge

🎯 Goal

Program a Scratch "robot" (sprite) to navigate through a maze and reach the finish without touching the walls.

🏗️ Setup

1. Open Scratch.
2. Draw or import a **maze background** (white path, black walls).
    - You can use the Scratch editor's "Paint" tool or find premade maze backdrops.
3. Add a **robot sprite** (any character).
4. Add a **goal/finish sprite** (like a star, flag, or portal).

🔧 Challenge Instructions

Students must:

1. **Movement**
    - Program the robot to move **forward, left, right** with arrow keys.
2. **Collision Detection**
    - If the robot touches a wall (black), it should:
        - Bounce back OR return to start.
3. **Victory Condition**
    - If the robot touches the goal sprite, display **"You Win!"**.

💡 Extensions for Extra Challenge

1. **Autonomous Navigation**
    - Instead of manual arrow keys, program the robot with logic (e.g., "If touching wall → turn right, else → move forward").
    - This makes it a **self-driving maze solver**.
2. **Timer & Score**
    - Add a timer variable.
    - Challenge students to finish the maze in the fastest time possible.
3. **Multiple Levels**
    - Create more than one maze and switch backdrops when the goal is reached.
4. **Sensors Simulation**
    - Have the robot use Scratch's **color sensing** (e.g., "if touching black → turn left").

- This mimics how a real robot would use line/edge sensors.

👩‍🏫 Teacher’s Prompt

"Your robot has been dropped into a maze. It can’t climb walls, but it can sense when it hits one. Program your robot to find its way to the goal. First, try controlling it with arrow keys. Then, see if you can make it solve the maze automatically without your help."

🟦 Sample Scratch Script (Autonomous Maze Robot)

Assumptions:

- Maze background = **white paths, black walls**
- Robot sprite starts at the entrance
- Goal = separate sprite (star/flag/etc.)

🔧 Step 1: Movement Basics

Use a loop to keep the robot moving forward a little at a time:

```
when green flag clicked
go to start position
point in direction 90
forever
   move 2 steps
end
```

🔧 Step 2: Wall Detection

Add logic so the robot "bounces" when hitting a black wall:

```
forever
   move 2 steps
   if touching color [#000000] then  // black wall
      move -2 steps    // back up
      turn clockwise (15) degrees
   end
end
```

This makes the robot "wiggle" along the maze until it finds a path.

🔧 Step 3: Goal Detection

Add a check for the finish:

```
if touching [Goal v] then
   say "I made it!" for 2 seconds
   stop all
end
```

🔧 Full Combined Script

```
when green flag clicked
go to x:-200 y:-150   // maze entrance
point in direction 90
forever
   move 2 steps
   if touching color [#000000] then   // wall color
      move -2 steps
      turn clockwise (15)
   end
   if touching [Goal v] then
      say "I made it!" for 2 seconds
      stop all
   end
end
```

```
when [flag] gnn clicked
go to x: -200 y: -150
point in directio 90
forever
  move 2 steps
  if <touching color [black]> then
    move -2 step
    turn clockwise 15
    if <touching [star] goal>
      say I made it! for 2 seconds
      stop all
```